

S1 Champion System — Overview, Terminology, UI Guide, and Performance Record

2026-08-23

S1 Champion System — Overview

As of 2026-08-23. This document describes the currently deployed Prediction Lab system end to end: the strategy itself, every term used around it, the infrastructure that runs it (paper trader, live executor, monitor), what each element of the monitoring UI shows and how to read it, and the performance record to date. It is a reference document; the authoritative registrations live in `data/predlab/gates.json` and the design documents under `docs/superpowers/specs/`.

1. What the system is, in one paragraph

The champion is a **daily-rebalanced, market-neutral long-short portfolio of cryptocurrency perpetual futures** built on a single anomaly: within the most liquid ~200 USDT-perpetual contracts on Binance, coins with **low recent price volatility** have tended to outperform coins with high recent volatility. Each day the system ranks the universe by a smoothed volatility estimate, buys the calmest quintile (40 names), shorts the most volatile quintile (40 names), equal weight on both sides, and scales the whole book so the portfolio targets 15% annualized volatility. No machine learning, no LLMs, no prediction of returns — the signal is a ranking of realized volatility. The system was selected through a pre-registered research program (Phase O, ~90+ trials, deflated-Sharpe accounting) and is now in a **sealed forward test**: a paper trader journals the book daily on a VPS, a live executor mirrors that book (currently on Binance testnet, real \$3,000 planned), and a web monitor displays both.

2. Strategy definition

2.1 Universe

- All Binance **USDT-margined perpetual futures** with contract status TRADING (~400+ symbols at any time).
- Each calendar month, membership is frozen: the **top 200 symbols by median daily dollar volume over the prior month** (a ~35-day pre-month window). Using the *prior* month prevents the ranking from peeking at the month being traded

(point-in-time discipline), and monthly freezing keeps turnover from membership churn low.

2.2 Signal — ewma_20 of Parkinson variance

- For each symbol and day, compute the **Parkinson variance estimate** $\ln(\text{high}/\text{low})^2 / (4 \cdot \ln 2)$ from the daily candle. The Parkinson estimator uses the intraday high-low range, which is a more efficient volatility proxy than close-to-close returns.
- Smooth it with an **exponentially weighted moving average with span 20** (ewma_20): recent days dominate, weights decay geometrically, ~20-day effective memory.
- The smoothed value is the signal: **low = attractive (long), high = unattractive (short)**.

2.3 Portfolio construction — quintile long-short

- Rank the 200 members by the signal. **Long the lowest quintile (40 names), short the highest quintile (40 names)**.
- **Equal weight** within each side: every long +2.5% of one book unit, every short -2.5%. Sides sum to +1 and -1, so the book has **gross exposure 2.0** (2× capital deployed across both sides) and net exposure ≈ 0 (market-neutral by construction — a broad market move hits longs and shorts alike).
- Rebalanced **daily at the UTC close**. Realized one-way turnover averages ~0.22 book units/day for the champion book (most names stay in their quintile day to day).

2.4 Overlay — vt15_b100 (volatility targeting + breadth floor)

The raw book's volatility drifts with the market. The overlay resizes it:

- **Vol target 15%**: $\text{scale} = 0.15 / \text{realized_vol}$, where realized_vol is the annualized standard deviation of the book's last 20 **gap-free** daily returns. If the book has been running hot at 68% annualized vol, $\text{scale} \approx 0.22$ — only ~22% of capital-equivalent exposure is deployed.
- **Scale cap 2.0**: leverage never exceeds 2× the base book even in dead-calm regimes.
- **Warm-up**: the scale is undefined (null) until **21 consecutive gap-free daily returns** exist. Until then the paper journal records the unscaled book and any live trading waits.
- **Breadth floor 100**: if fewer than 100 universe members have a valid signal on a given day (a data-quality symptom), the scale is forced to 0 — the book goes flat rather than trade a degraded universe.

2.5 Costs

All backtest and forward accounting assumes **taker fees** on every unit of turnover (TAKER_BP, ~5 bp per side). At ~0.22 daily turnover this costs ~1 bp/day. Fill real-

ism beyond the fee (queue, impact, timing) is exactly what the live executor phase measures.

2.6 Parameter summary

Parameter	Value
Universe	Top 200 Binance USDT-perps, prior-month median dollar volume, monthly frozen
Signal	ewma(span 20) of daily Parkinson variance
Book	Quintile long-short, 40L/40S, equal weight, gross 2.0
Rebalance	Daily, UTC close
Overlay	Vol target 15%, 20-day gap-free window, scale cap 2.0, breadth floor 100
Costs	Taker fees on turnover

3. Terminology glossary

Term	Meaning
Parkinson variance	Volatility estimate from the daily high-low range: $\ln(H/L)^2 / (4 \ln 2)$. More statistically efficient than squared close-to-close returns.
EWMA (span 20)	Exponentially weighted moving average; recent observations weighted most, ~20-day effective memory.
Quintile	One fifth of the ranked universe (here 40 of 200 names).
Gross / net exposure	Gross = sum of absolute position sizes (2.0 here). Net = longs minus shorts (≈ 0 here $\rightarrow$ market-neutral).
Vol targeting	Scaling position size so the portfolio's realized volatility tracks a target (15%). Cuts size after volatile stretches, adds after calm ones.
Scale (vt15_b100_scale)	The overlay multiplier actually applied: 0.15/realized vol, capped at 2.0, null during warm-up, 0.0 when the breadth floor trips.
Breadth	Number of universe members with a valid signal on the day.

Term	Meaning
realized_book_ret	The day's raw book return (at gross 2.0, unscaled), computed close-to-close from the previous journal row's weights. Not an account return.
NAV / account return	Return on total capital: $\text{scale} \times \text{realized_book_ret}$, compounded. This is the number that answers "if \$10k is on the account and it made \$1k, show +10%."
mark_px / realized_mark_ret	Prices snapshotted at the moment the journal row is written (minutes after the UTC close), and the book return measured between those snapshots. The difference vs the close-to-close return exposes what the "fills at the close" assumption hides.
Slippage (bp)	Execution shortfall vs an assumed price, in basis points (1 bp = 0.01%). Positive = cost.
Sharpe ratio (SR)	Annualized mean return divided by annualized volatility. The headline risk-adjusted performance number.
Deflated Sharpe Ratio (DSR)	Probability that the observed SR is not an artifact of testing many strategies: corrects for the number of trials, non-normality, and track length. DSR 0.913 = 91.3% confident the edge is real after multiplicity correction.
Placebo test	Re-running the backtest with the causal link deliberately broken (e.g. time-shifted signals). A strategy that still "works" under placebo is exploiting an artifact.
Sealed one-shot forward test	Pre-registered evaluation: criteria and date fixed in <code>gates.json</code> <i>before</i> any forward data existed. One evaluation, earliest 2027-01-02; no peeking-and-adjusting.
Pre-registration	Writing the hypothesis, method, and pass/fail gates into <code>gates.json</code> before running the experiment, so results cannot be cherry-picked after the fact.
Journal-follower	The live executor's design: it never computes signals; it reads the book the paper trader already journaled and reproduces it on the exchange. Signal parity is guaranteed by construction.

Term	Meaning
Reduce-only order	Exchange order flag: may only shrink an existing position, never open or grow one. Used for all closes and the emergency flatten.
minNotional / stepSize	Per-symbol exchange minimums: smallest order value (typically 5-50 USDT) and quantity granularity. Legs too small to clear them are dropped and logged.
Implementation shortfall	Total live-vs-paper performance gap: fees + slippage + dropped legs + timing. The quantity the live phase measures.
Halt flag	A file (<code>halt.flag</code>) whose presence blocks all executor trading. Written automatically on a daily-loss breach or manually via <code>close-all</code> ; only manual removal resumes trading.
Warm-up (21 days)	Days of gap-free returns required before the vol-target scale is defined.

4. Infrastructure

Everything runs on the Hetzner VPS (46.225.169.184); the laptop runs nothing.

4.1 Paper trader (`scripts/predlab_s1_paper.py`) — the registered forward test

- Hourly cron (minute 15); the first wake after the UTC day rolls does the work, later wakes exit cheaply (idempotent).
- Fetches daily candles for all perps from the public Binance API, rebuilds the book for the last complete UTC day, and appends one row per day to two journals under `predlab-data/predlab/s1_paper/`:
 - `journal_champion.jsonl` — the Phase-O champion (`ewma_20`, `vt15_b100`). **This is the sealed forward test record.**
 - `journal.jsonl` — the older S1 book (`park_5` signal, `vt10` overlay), kept for a separate registered confirmation.
- Each row: date, weights (40L/40S), realized return of yesterday's book, turnover and cost estimates, the vol-target scale, breadth, and the write-time price snapshot (`mark_px`) enabling the fill-assumption check.
- Rows are append-only and idempotent; journals are backed up daily (00:45 UTC) to a git branch. **No orders are placed by this component.**

4.2 Live executor (`scripts/predlab_s1_live.py`) — the measurement instrument

Purpose: measure what real execution costs. It chains after the paper cron and:

1. Reads the latest champion journal row (never recomputes signals).

2. Safety gates, in order: halt flag → daily-loss check (equity < 95% of day-start → flatten everything reduce-only + halt) → idempotency (day already executed → exit) → scale defined? (null → WAIT) → marks present? → hedge-mode check (account must be in one-way position mode).
3. Sizes each leg: $\text{weight} \times \min(\text{scale}, 1.1) \times \text{account equity}$. The **1.1 clamp** exists because the overlay's own cap (2.0) would exceed the executor's risk rails; on days the overlay wants more, the live book is a proportionally smaller replica (recorded as `scale_raw` vs `scale` in the journal).
4. Diffs targets against actual exchange positions and places **market orders** for the deltas. Positions whose symbol left the champion book are always closed in full. Legs below exchange minimums are dropped and logged.
5. Risk rails (hard): $\text{gross} \leq 2.2 \times \text{equity}$, per-symbol $\leq 5\%$ of gross, leverage 4, cross margin, 5% daily-loss halt, `halt.flag` kill switch, `close-all` emergency command.
6. Writes its own journals (`journal_live.jsonl`, `fills.jsonl`) with every fill's price and fee — the raw material for the slippage measurement.

Rollout (current position in bold):

Phase	What	Status
Dry-run	Full pipeline, zero orders	done
Testnet rehearsal	Real orders on Binance <i>testnet</i> (--testnet, separate data dir). Plumbing validation only — testnet books are thin and fills there feed no conclusions.	armed; first cycle expected 2026-08-25 ~00:15 UTC (waiting for the 21-day scale warm-up: 19/21 as of Aug-23)
Live	\$3,000 real USDT, after ≥ 2 clean testnet days and an explicit go/no-go	pending

The live run is registered in `gates.json` (`predlab_s1_live`) as **observational** — a measurement, not a pass/fail experiment; the sealed forward test is unaffected by anything the executor does.

4.3 Monitor (web UI, live-v2.7.2)

FastAPI backend + React frontend served from the VPS behind password auth. Reads the journals read-only every 30 s. Described in §5.

5. What the UI shows and how to read it

The monitor's Prediction Lab section has four tabs. Two books appear throughout: **champion** (the system described above) and **vt10** (the older `park_5` book — context only).

5.1 Performance tab

- **Equity curves (base 100)** — each book’s raw daily returns compounded from 100. **This is the unscaled book at gross 2× exposure, before the overlay** — it answers “is the signal working?”, not “what would an account have made?”. The card is labeled “*Book return (gross 2× unscaled)*” for exactly this reason.
- **Account NAV (scaled)** — *new*. The overlay-adjusted, account-percent cumulative return: each day contributes $\text{scale} \times \text{book return}$, where the scale used is the one that was known when the position was put on. **This is the number that scales to portfolio size** (\$10,000 account, +10% shown → \$1,000 made). Until the 21-day warm-up completes it shows “*overlay warming up n/21*” — no scale existed, so no scaled return is claimable; the series starts accruing 2026-08-25.
- **Account cards (testnet / live)** — *new*. Appear once the executor trades: actual account equity in USDT, cumulative % vs. starting equity (fees and slippage included — the truest performance number), cycle count, a **HALTED** badge if the halt flag is present, and a (*dry-run*) suffix where applicable. The gap between the paper NAV curve and this curve **is** the implementation shortfall being measured.
- **Fill slippage card** — mean and cumulative difference (bp) between the close-to-close return the journal assumes and the mark-to-mark return actually observable at write time. Negative mean = the close-fill assumption has been flattering the paper record. Only days since 2026-08-18 carry marks.
- **Rolling Sharpe (30d) and drawdown series** — risk-adjusted trend and peak-to-trough loss of the raw book.
- **Reference / backtest yearly** — the frozen dev-period metrics and yearly overlay returns from the registered backtest, for eyeballing forward vs. backtest.

5.2 Book tab

Today’s portfolio composition: the 40 longs and 40 shorts with weights, universe size, **breadth** (signal coverage; the floor is 100), **membership hash** (fingerprint of the name list — parity checks against the executor), estimated turnover and cost for the day, and how many names entered/exited vs. yesterday.

5.3 Gate tab

The sealed one-shot forward test tracker — **informational only**; the actual evaluation stays sealed:

- Window started 2026-07-02; earliest evaluation **2027-01-02**; days elapsed/remaining.
- Pass criteria, fixed in advance: net overlaid forward SR ≥ 0.946 (half the dev SR of 1.892), same sign as dev, time-shift placebo $p < 0.10$, one evaluation only.
- A running SR proxy from the paper journal — a progress glance, explicitly *not* the official evaluation (which uses the backtest harness on the sealed window).

5.4 Ops / Health tab

Freshness and integrity: last row age per journal (stale after 36 h), row and malformed-line counts, calendar gaps (the 2026-07-31...08-02 VPS migration gap is marked as known), and the daily journal-backup heartbeat note.

6. Performance record to date

6.1 Registered backtest (development, pre-registered; 2020-2026 data)

Metric	Value
Overlay net Sharpe (full period)	+1.892
Max drawdown (overlay)	17.6%
Deflated Sharpe Ratio	0.913 ($\geq 90\%$ confidence after $\sim 90+$ trial correction)
Yearly results	6 of 6 years positive
Phase-P S1 holdout (sealed, one-shot)	net SR +2.20 — passed

Supporting validation:

- **Venue replication (Bybit, 735 symbols):** champion applied verbatim on a different exchange's data — raw SR +1.94 / overlay +1.71, placebo clean, all folds positive. Robustness check (same market, so not an independent sample).
- **Capacity study:** square-root impact model puts the strategy at SR ≈ 1.0 around **\$21M** deployed and SR ≈ 0 around \$91M; the binding constraint is short-leg small-caps. At \$3K, capacity is a non-issue.
- **Cross-asset probes (equities, futures/FX):** the same recipe applied outside crypto perps is flat-to-negative — the edge is a crypto-perp phenomenon, which bounds how far the result generalizes.

6.2 Forward paper record (the sealed window, so far)

As of 2026-08-22 (19 return days for the champion):

	Champion (ewma_20)	vt10 book (park_5)
Journal span	2026-08-03 → 08-22	2026-07-30 → 08-22
Cumulative return (raw book, gross 2 $\times$, unscaled)	+34.7%	+21.9%
Mean daily return	+183 bp	+110 bp
Annualized vol (raw book)	68.5%	60.6%
Naive annualized SR	+9.7	+6.6
Mean daily turnover / cost	0.22 / 1.1 bp	0.65 / 3.3 bp
Mark-to-mark check (4 days so far)	+19.6%	+6.2%

Read these numbers with three caveats:

1. **They are unscaled book returns.** The overlay would have deployed only $\sim 0.2\times$ (15% target $\div \sim 68\%$ realized vol), so the account-equivalent ("NAV") return over the window is roughly **+7%**, not +34.7%. The UI's new NAV card exists precisely to prevent this misreading.

2. **19 days is a tiny sample in an unusually good stretch.** An audit placed this window around the **87th percentile** of the historical mean-return distribution. The anchored expectation remains the backtest's overlay $SR \approx 1.9$, not 9.7.
3. **Costless fills assumed.** The close-to-close journal assumes frictionless fills at the close; the mark-to-mark leg and, from Aug-25, real testnet and then real-money fills progressively replace that assumption with data.

6.3 Live execution status

- Executor deployed and armed on the VPS; every hourly wake currently exits WAIT (scale warm-up 19/21). First testnet trading cycle expected **2026-08-25 ~00:15 UTC**; the testnet account's 7 leftover positions will be flattened on that first cycle (an incidental live test of the close path).
- Success criteria for the testnet phase: ≥ 2 consecutive clean days (journal row each day, zero unexplained order errors, positions matching targets for symbols listed on testnet). Then a go/no-go decision gates the \$3,000 live phase, whose watch procedure is in docs/predlab/sl_live_runbook.md.

7. Governance summary

Every claim above is bounded by the pre-registration discipline: the champion's configuration and pass criteria were frozen in gates.json before the forward window opened; the forward evaluation is one-shot and sealed until 2027-01-02; the live run is registered as observational and cannot be promoted into evidence for the gate; and the raw-vs-scaled distinction in the UI keeps the headline numbers honest. The corresponding thesis sections are §54–§61.